LogNormal

Parametrisation

The LogNormal has density

$$f(y) = \frac{1}{y\sqrt{2\pi}}\sqrt{\tau}\exp\left(-\frac{1}{2}\tau(\log y - \mu)^2\right), \qquad y > 0$$

where

$\tau > 0$ is the precision parameter,

μ is the mean parameter.

Link-function

The parameter μ is linked to the linear predictor as:

$$\eta = \mu$$

Hyperparameters

The τ parameter is represented as

$$\theta = \log\tau$$

and the prior is defined on θ.

Specification

- `family="lognormal"` for regression models and `family="lognormalsurv"` for survival models.
- Required arguments: y. Given in a format by using `inla.surv()` function for `"lognormal.surv"`

Hyperparameter spesification and default values

lognormal

doc `The log-Normal likelihood`
hyper
theta
hyperid `77101`
name `log precision`
short.name `prec`
output.name `Precision for the lognormal observations`
output.name.intern `Log precision for the lognormal observations`
initial `0`
fixed `FALSE`
prior `loggamma`
param `1 5e-05`
to.theta `function(x) log(x)`
from.theta `function(x) exp(x)`
survival `FALSE`
discrete `FALSE`
link `default identity`
pdf `lognormal`

lognormalsurv

doc The log-Normal likelihood (survival)

hyper

theta1

hyperid 78001

name log precision

short.name prec

output.name Precision for the lognormalsurv observations

output.name.intern Log precision for the lognormalsurv observations

initial 0

fixed FALSE

prior loggamma

param 1 5e-05

to.theta function(x) log(x)

from.theta function(x) exp(x)

theta2

hyperid 78002

name beta1

short.name beta1

output.name beta1 for logNormal-Cure

output.name.intern beta1 for logNormal-Cure

initial -7

fixed FALSE

prior normal

param -4 100

to.theta function(x) x

from.theta function(x) x

theta3

hyperid 78003

name beta2

short.name beta2

output.name beta2 for logNormal-Cure

output.name.intern beta2 for logNormal-Cure

initial 0

fixed FALSE

prior normal

param 0 100

to.theta function(x) x

from.theta function(x) x

theta4

hyperid 78004

name beta3

short.name beta3

output.name beta3 for logNormal-Cure

output.name.intern `beta3 for logNormal-Cure`
initial `0`
fixed `FALSE`
prior `normal`
param `0 100`
to.theta `function(x) x`
from.theta `function(x) x`

theta5
hyperid `78005`
name `beta4`
short.name `beta4`
output.name `beta4 for logNormal-Cure`
output.name.intern `beta4 for logNormal-Cure`
initial `0`
fixed `FALSE`
prior `normal`
param `0 100`
to.theta `function(x) x`
from.theta `function(x) x`

theta6
hyperid `78006`
name `beta5`
short.name `beta5`
output.name `beta5 for logNormal-Cure`
output.name.intern `beta5 for logNormal-Cure`
initial `0`
fixed `FALSE`
prior `normal`
param `0 100`
to.theta `function(x) x`
from.theta `function(x) x`

theta7
hyperid `78007`
name `beta6`
short.name `beta6`
output.name `beta6 for logNormal-Cure`
output.name.intern `beta6 for logNormal-Cure`
initial `0`
fixed `FALSE`
prior `normal`
param `0 100`
to.theta `function(x) x`
from.theta `function(x) x`

theta8
hyperid `78008`

name beta7
short.name beta7
output.name beta7 for logNormal-Cure
output.name.intern beta7 for logNormal-Cure
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta9
hyperid 78009
name beta8
short.name beta8
output.name beta8 for logNormal-Cure
output.name.intern beta8 for logNormal-Cure
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta10
hyperid 78010
name beta9
short.name beta9
output.name beta9 for logNormal-Cure
output.name.intern beta9 for logNormal-Cure
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta11
hyperid 78011
name beta10
short.name beta10
output.name beta10 for logNormal-Cure
output.name.intern beta10 for logNormal-Cure
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x

from.theta `function(x) x`

survival `TRUE`

discrete `FALSE`

link `default identity`

pdf `lognormal`

Example

In the following example we estimate the parameters in a simulated case

```
n = 300
x = c(scale(runif(n)))
eta = 1+2.2*x
y = exp(rnorm(n, mean = eta,  sd = 1))
data = list(y=y, event=rep(1, n), x=x)
formula = inla.surv(y, event) ~ 1 + x
r=inla(formula, family ="lognormalsurv", data=data)
summary(r)

data = data.frame(y, x)
formula = y  ~ 1 + x
r=inla(formula, family ="lognormal", data=data)
summary(r)
```

Notes

- lognormalsurv can be used for right censored, left censored, interval censored data. A general framework to represent time is given by `inla.surv`. If the observed times y are large/huge, then this can cause numerical overflow, and if you encounter this problem, try to scale the observatios, like `time = time / max(time)`.